

FlowPilot: A Multi-Agent AI System for Continuous Flow Chemistry Process Design

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ABSTRACT

Translating a batch protocol into continuous flow requires chemical reasoning, engineering calculations, and equipment choices to remain consistent. We present FlowPilot, a multi-agent AI system connecting these decisions through standardized chemist intake, literature retrieval, deterministic calculation, specialist review, and inventory reconciliation. A corpus of 464 classified literature records and 2,537 handbook-derived rules supports the workflow. A common design record links stage conditions, stream assignments, equipment, and the process diagram. Matched one-shot and full-pipeline generation across three chemistry cases, five retained generator models, and three repeats yields 90 outcomes. Separate LLM judges apply 14 fixed criteria informed by deterministic checks. The mean benchmark score increases from 0.794 to 0.917, while raw judge critical flags decrease from 98 to 4. Laboratory implementation examines two connected-stage reactions: photoredox Giese addition followed by oxidation affords a sulfoxide in 97–98% NMR yield, and DPDTC-mediated amidation affords an amide in 68–86% NMR yield. Selected conditions give isolated yields of 94% and 84%, respectively. These results connect computational design consistency with experimentally demonstrated synthesis, while distinguishing proposed settings, implemented operation, and measured chemical outcomes.

INTRODUCTION

Continuous-flow chemistry offers a versatile platform for synthesis by controlling how reagents are mixed, heated, irradiated, and brought into contact with gases or catalysts.¹⁻⁴ These capabilities have

expanded access to photochemical, electrochemical, and gas-liquid transformations and enabled practical routes to pharmaceutical intermediates.⁵⁻¹⁰ Connecting reaction stages can further reduce intermediate handling and support multistep or on-demand production.¹¹⁻¹⁴ Realizing these advantages, however, requires more than transferring a batch recipe into a smaller vessel. A successful translation must connect the chemistry to a suitable reactor arrangement, quantitative feed conditions, and the equipment available in the laboratory.^{15,16}

Those choices are closely coupled. Reactor geometry influences heat and mass transfer, irradiation, and pressure loss, while an interstage addition changes both composition and the flow entering the next reactor.¹⁷⁻²¹ Gas-liquid processes introduce a further distinction between gas molar delivery and its pressure-dependent volume, making the definition of residence time particularly important.²² In a telescoped sequence, the composition leaving one stage also determines what the next stage can achieve. Integrated apixaban synthesis and photochemical routes to sartan intermediates illustrate the coordination of chemistry, hardware, and operating conditions required in multistep flow development.^{23,24}

Digital methods increasingly support this development process. Computer-aided synthesis planning identifies reaction routes,²⁵⁻²⁷ process simulation evaluates connected unit operations,^{28,29} and Bayesian and active-learning methods guide experimental optimization.³⁰⁻³⁴ AI-informed robotic flow synthesis has linked planning to physical execution,³⁵ while RoboChem-Flex and UV-visible auto-reactometry extend modular optimization and LLM-assisted photochemical screening.^{36,37} These approaches demonstrate the value of integrating computation with experimentation. Preparing an initial configuration from a conventional reaction procedure remains an important entry point, especially when a laboratory must work within a specific set of pumps, reactors, and control modules.

Large language models provide an interface between such procedural descriptions and computational tools. ChemCrow and Coscientist combine chemical reasoning with tool use and experimentation, and El Agente coordinates computational chemistry workflows.³⁸⁻⁴⁰ Co-Scientist and Paper2Agent extend collaborative scientific reasoning and the reuse of computational methods.^{41,42} For laboratory procedures, ACRA links literature instructions to executable chemical descriptions and available hardware; AutoLabs combines clarification, calculation, and self-correction for liquid handling; and PRISM incorporates simulation feedback before robotic execution.⁴³⁻⁴⁵ Together, these studies show how language-based interaction can connect scientific intent to structured operations.

Several systems address flow chemistry directly. Chat-microreactor uses literature-derived information for flow-pattern prediction and microreactor design guidance, including a piperacillin application.⁴⁶

SapoMind combines condition recommendation with experimental optimization of continuous-flow lanolin saponification.⁴⁷ Ruan et al. developed LLM-RDF for end-to-end synthesis development, integrating experimental execution, kinetic analysis, optimization, purification, and reactor-design tasks.⁴⁸ These studies establish both quantitative engineering and experimental development as part of the emerging scope of chemical agents. For batch-to-flow translation, the practical task is to bring these decisions together in a connected process whose feed compositions, reactor conditions, and equipment assignments can be inspected as a whole.

Related work in chemical process design provides complementary approaches to this integration. Tian et al. generate process topologies and parameters for executable simulation flowsheets,⁴⁹ and the CeProAgents preprint organizes knowledge extraction, conceptual design, and parameter optimization into specialist agent groups.⁵⁰ The CAAF preprint investigates deterministic constraint checking and candidate revision, including a constructed flow-reactor design problem.⁵¹ The El Agente Gráfico and La Agente Óptima preprints from the Aspuru-Guzik group further explore structured execution and agent-supervised optimization.^{52,53} FlowPilot builds on this broader direction while focusing on the relationship between a reaction procedure and its laboratory-specific flow configuration: which streams enter each stage, how their compositions and flow rates determine operating conditions, and which available instruments can realize the connected design. **Table S36** provides the detailed literature comparison.

Assessing this capability requires more than fluent explanations or plausible individual settings. Chemical reasoning benchmarks have identified elementary errors and overconfidence in otherwise capable models,⁵⁴ and uncertainty-aware optimization studies have documented invalid suggestions within defined reaction search spaces.⁵⁵ For a flow process, the corresponding concern is whether the chemical sequence, material balance, reactor geometry, and operating procedure agree. Evaluation therefore needs to examine the consistency of the complete proposed process as well as its eventual chemical performance.

Here we present FlowPilot, a batch-to-flow design workflow that combines standardized chemist intake, chemistry-aware literature retrieval, deterministic engineering calculations, specialist council review, and inventory assignment. A shared final design record links the stream compositions and flow rates to stage conditions, equipment selections, and the process diagram. We evaluate retrieval alignment, compare delivered designs with matched one-shot generation using fixed criteria and separate LLM judges, and examine the effects of council components and candidate exploration.⁵⁶ Two laboratory implementations then test complementary aspects of connected-process design: photoredox Giese addition followed by oxidation requires staged oxygen introduction (**Figure 5**), whereas DPDTC-mediated amidation requires interstage reagent addition and transfer of a reactive intermediate without isolation (**Figure 6**).^{57,58}

RESULTS

FlowPilot organizes batch-to-flow translation around a shared process description (**Figure 1**). The chemist supplies the protocol, design objective, and laboratory inventory, then resolves standardized follow-up questions about missing chemistry, constraints, and experimental history. The confirmed answers form a stored input package for the run. The upstream module converts that context into a chemistry plan identifying species, reaction stages, stream incompatibilities, sensitivities, and possible flow-specific advantages. Retrieval supplies relevant precedents, and engineering calculations establish an initial proposal containing reactor geometry, stream assignments, residence-time assumptions, and operating conditions. Council review and equipment reconciliation then assess that proposal before it becomes the final reported design.

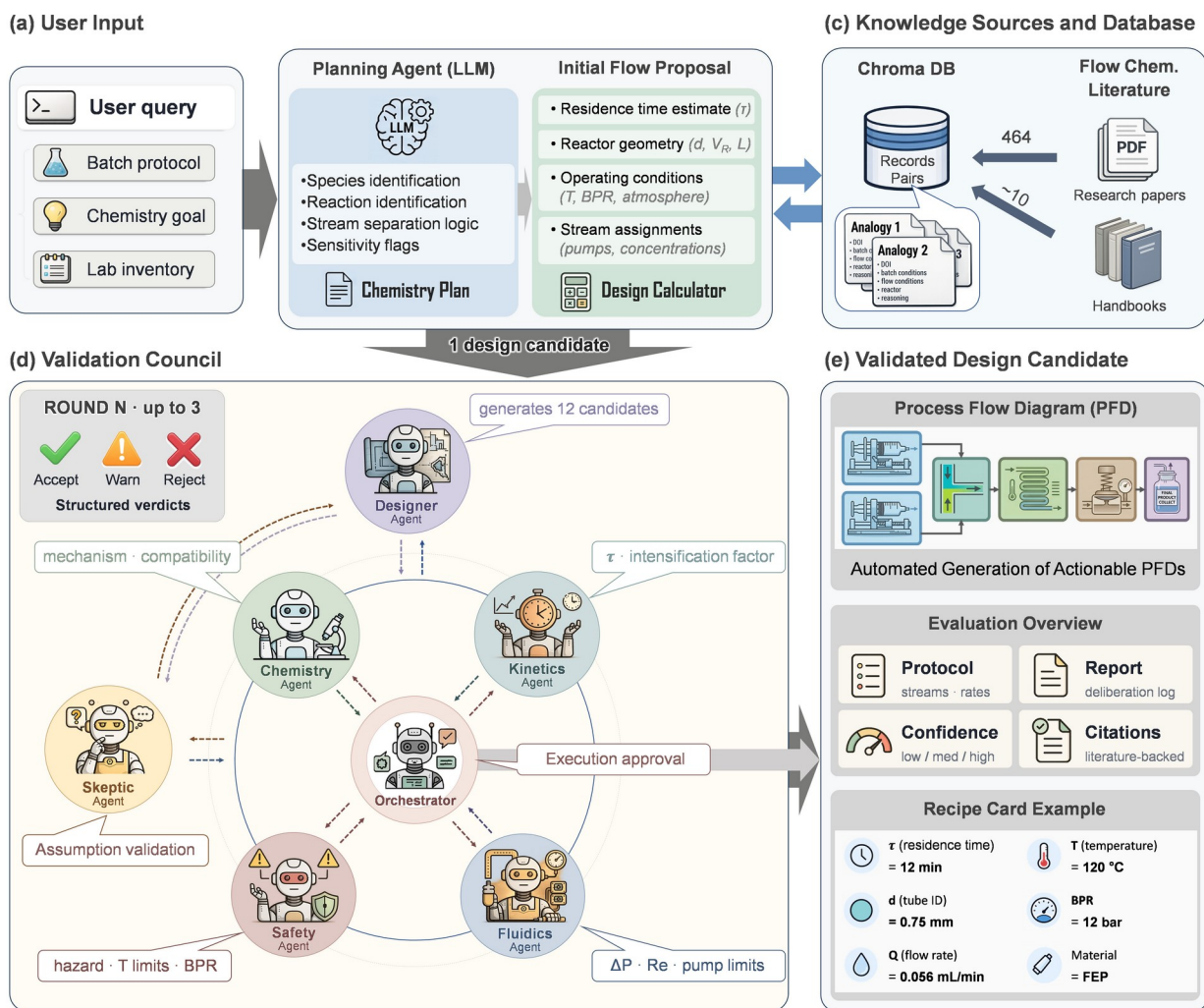


Figure 1. FlowPilot system architecture. User inputs are converted into a chemistry plan and an initial engineering proposal, supported by flow-chemistry literature and handbook knowledge. The seven-agent council generates and reviews candidate designs, records structured verdicts, and proposes bounded revisions. The selected candidate is reconciled with engineering calculations and laboratory inventory before a process-flow diagram, protocol, report, confidence assessment, and supporting citations are returned. The illustrated recipe card is schematic; computational validation does not establish experimental performance.

The downstream council supplies chemical and cross-domain review alongside deterministic numerical checks (**Figure 1d**). Rather than asking one model to produce a single final answer, the system generates a candidate pool and distributes review across domain-specialized agents. The Designer proposes candidate geometries and operating points; Chemistry Agent evaluates chemical compatibility,

photochemical constraints, and stream logic; Kinetics Agent evaluates residence time and conversion plausibility; Fluidics Agent evaluates flow regime, pressure drop, mixing, and hardware compatibility; Safety Agent evaluates pressure, thermal, material, and handling risks; the Skeptic audits assumptions and internal consistency; and the Orchestrator selects the final candidate. This structure produces not only a selected design but also a record of rejected alternatives, revised candidates, disqualified candidates, and the rationale for each decision.

For a reconciled result, the output is a structured design package containing stage and stream parameters, equipment assignments, a process diagram, supporting analogies, and the review record. Numerical views and the diagram refer to the same final design contract, while unresolved requirements remain visible. This common record connects a chemical proposal to specific equipment and makes subsequent experimental feedback attributable to the conditions actually considered. The authority boundaries, input contracts, and provenance mechanisms are described in **Figures S1-S2** and **Tables S1-S3**; computational closure does not itself establish reaction performance.

Literature and rule-base coverage

The literature dataset covers a broad range of flow-chemistry applications (**Figure 2**). The curated export contains 464 classified literature records spanning thermal synthesis, photoredox catalysis, heterogeneous photocatalysis, cross-coupling, oxidation/reduction, and other reaction classes. Reactor types range from capillary/coil and microreactor/chip systems to photoreactors and packed beds, with fluoropolymer tubing prominent among the reported materials. Explicit inlet-stream counts are available for 154 processes; two- and three-stream configurations account for 91 and 30 of these records, respectively. These annotations supply precedents rather than prescribing a default number of streams. The diversity supports retrieval across photochemical and non-photochemical processes with different materials and operating regimes.

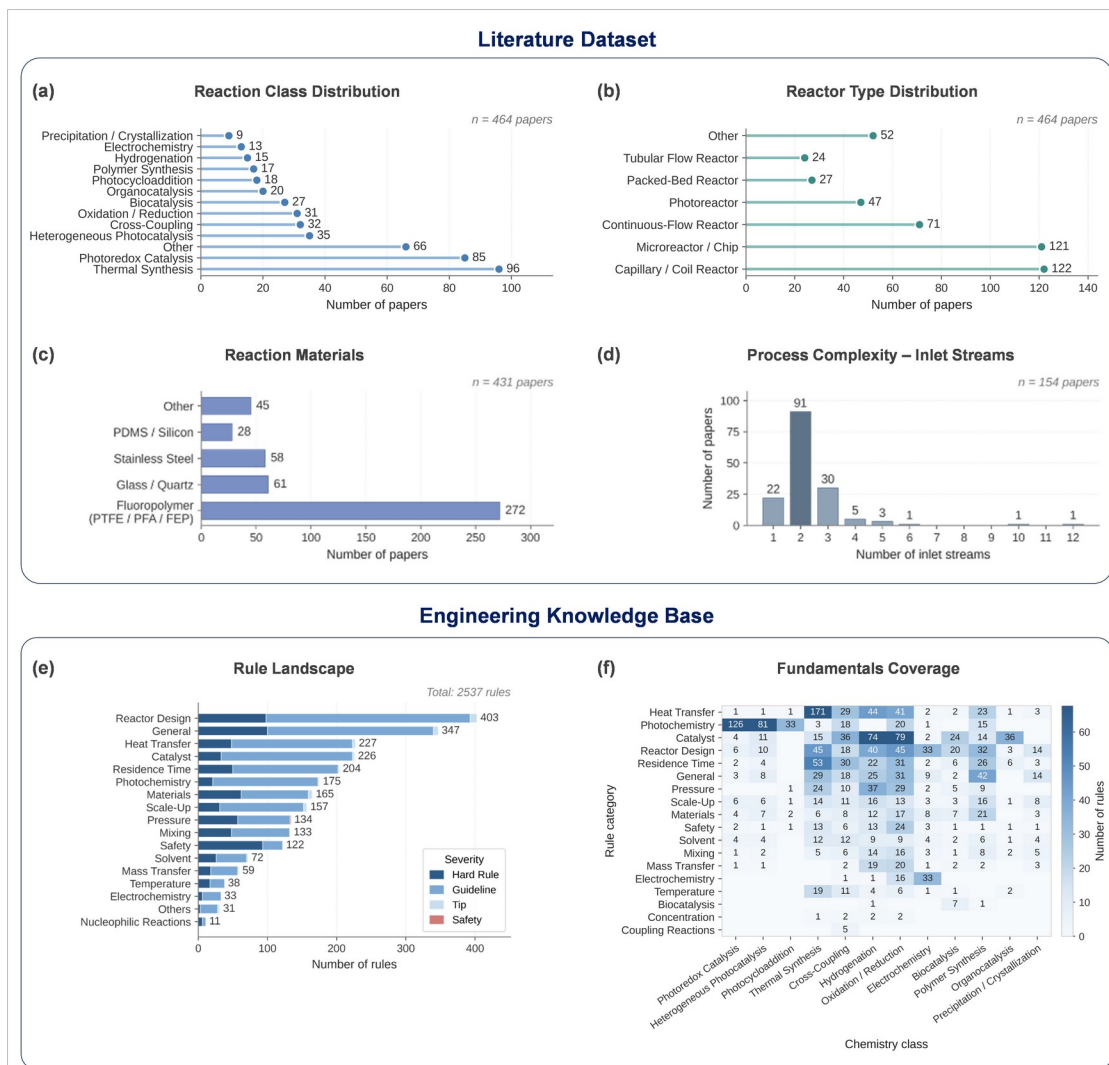


Figure 2. Literature corpus and engineering rule-base coverage. (a-d) Reaction classes, reactor types, reported materials, and inlet-stream configurations in the curated flow-chemistry corpus; explicit inlet-stream counts are available for 154 records. (e) Category and severity distribution within the 2,537-rule engineering knowledge base. (f) Recorded associations between rule categories and chemistry classes. Full categorical exports and field-specific availability are reported in **Figures S3-S4** and **Tables S4-S5**; associations indicate knowledge coverage rather than independent validation of every rule.

The engineering knowledge base contained 2,537 structured rules spanning reactor design, heat transfer, catalyst behavior, residence time, photochemistry, materials, scale-up, pressure, mixing, and safety (**Figure 2e-f**). Coverage was not uniform across chemistry classes: heat-transfer rules were densely associated with thermal synthesis and cross-coupling, whereas photochemistry rules concentrated in photoredox and photocycloaddition classes. This structured rule landscape supplies domain context for

evaluating model-generated designs alongside deterministic checks, rather than relying only on model priors. The frozen corpus composition and computed rule-base coverage are reported in **Figures S3-S4** and **Tables S4-S5**.

Plan-aware retrieval improves analogy quality

Coverage alone does not show that a relevant precedent will be selected. We therefore evaluated the retrieval system against a semantic-search baseline (**Figure 3**). The baseline embeds the batch protocol or query summary and ranks records by cosine similarity. FlowPilot enriches the query using the chemistry plan, including reaction class, mechanism, photocatalyst family, solvent, and wavelength. Filtering starts with paired records under mechanism and phase constraints, relaxes filters when fewer than three matches remain, and expands to all records if no paired match remains. A field-aware score then reranks the candidates. **Figure 3a** summarizes the workflow; **Figure S5** and **Table S6** specify its tier transitions and scoring weights.

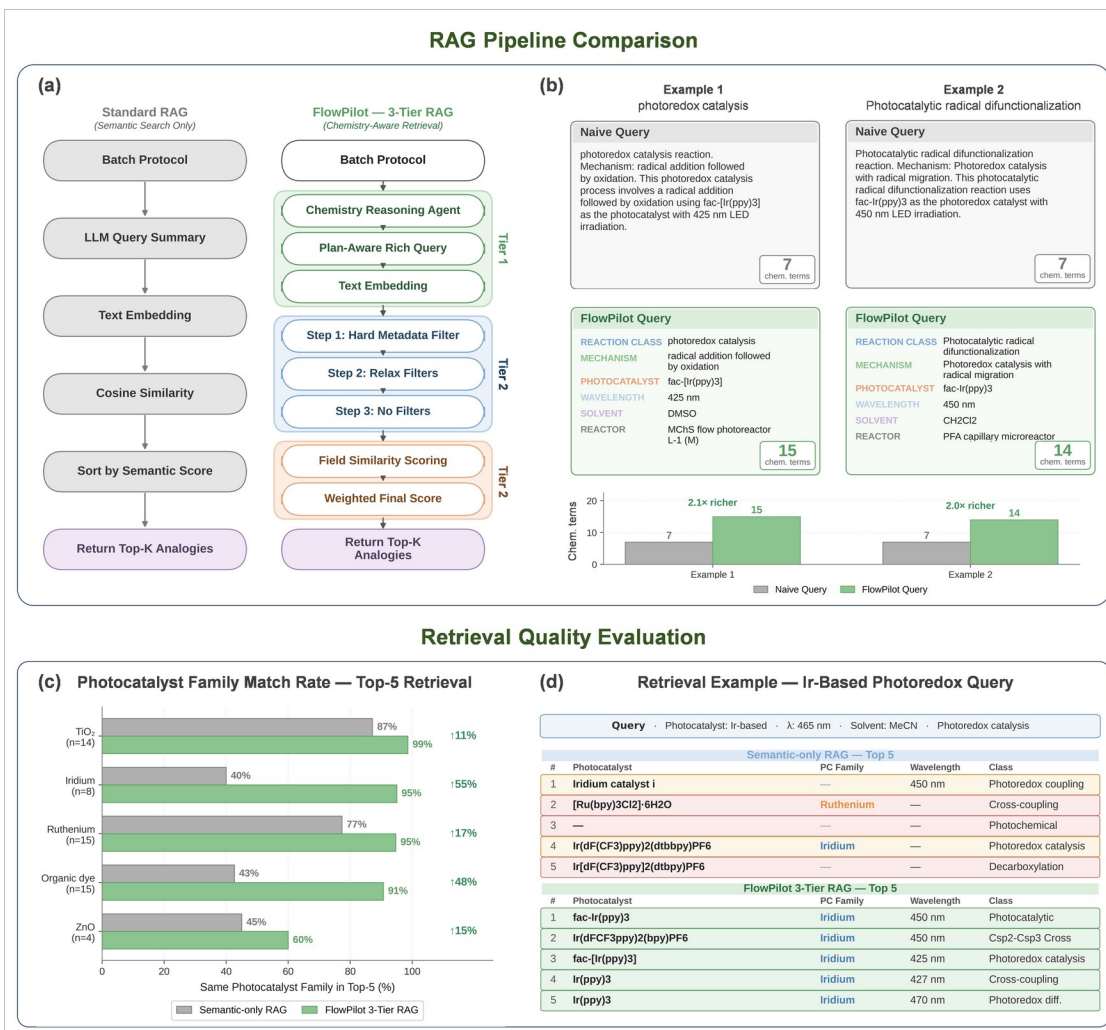


Figure 3. Plan-aware retrieval. (a) Three-tier filtering strategy: hard metadata filter, relaxed metadata filter, and pure semantic fallback. (b) Query enrichment expands chemically specific search terms approximately two-fold relative to a naive semantic baseline. (c) Top-5 photocatalyst-family matching rates by family (iridium, organic dye, ruthenium, TiO₂, ZnO) for semantic baseline vs FlowPilot reranking. (d) Representative reranking outcome showing closer wavelength and photocatalyst-family alignment.

The enriched query representation approximately doubled the number of chemically specific search terms in the two examples shown in **Figure 3b**. A photoredox-catalysis query increased from 7 naive terms to 15 FlowPilot terms, and a photocatalytic radical difunctionalization query increased from 7 to 14 terms. In the retrieval-pair benchmark, 1,600 query–result pairs were evaluated. FlowPilot reranked 25.1% of

these 1,600 pairs relative to the pure semantic baseline, indicating that field-aware scoring changed the ordering of the candidate analogies presented to the design engine.

The strongest retrieval improvement was observed for photocatalyst-family matching. For iridium systems, top-five family matching increased from 40.0% under semantic search to 95.0% after reranking. Organic-dye systems improved from 42.7% to 90.7%, ruthenium systems from 77.3% to 94.7%, TiO₂ systems from 87.1% to 98.6%, and ZnO systems from 45.0% to 60.0% (**Figure 3c**). The demonstration in **Figure 3d** illustrates how field-aware ranking promotes records with a matching photocatalyst family and relevant wavelength context. These results measure metadata alignment, not the accuracy of a transferred kinetic estimate or the eventual reaction yield. Retrieval tiers, scoring weights, benchmark data, and source-exclusion controls are detailed in **Figures S5-S6** and **Tables S6-S7**.

Ablation and architecture comparison

Improved retrieval does not by itself establish that the resulting flow design is consistent. We therefore compared one-shot generation with the full FlowPilot pipeline while holding the chemistry, inventory, and requested output contract constant (**Figure 4a**). The three cases were CuAAC, photochemical oxidation, and hydrogenolysis, with three generation repeats per model-architecture-case combination. Separate Qwen, OpenAI, and Claude judge calls evaluated normalized final designs against 14 fixed criteria, using the same source context and deterministic verification evidence. Criteria address chemical fidelity, stream and reactor consistency, equipment feasibility, operating procedure, and uncertainty. Applicable integer ratings from 0 to 4 were averaged with equal criterion and judge weights and normalized to a score from 0 to 1. Critical flags were recorded separately. **Tables S8-S10** provide the scoring anchors, criterion definitions, and module-screen summaries.

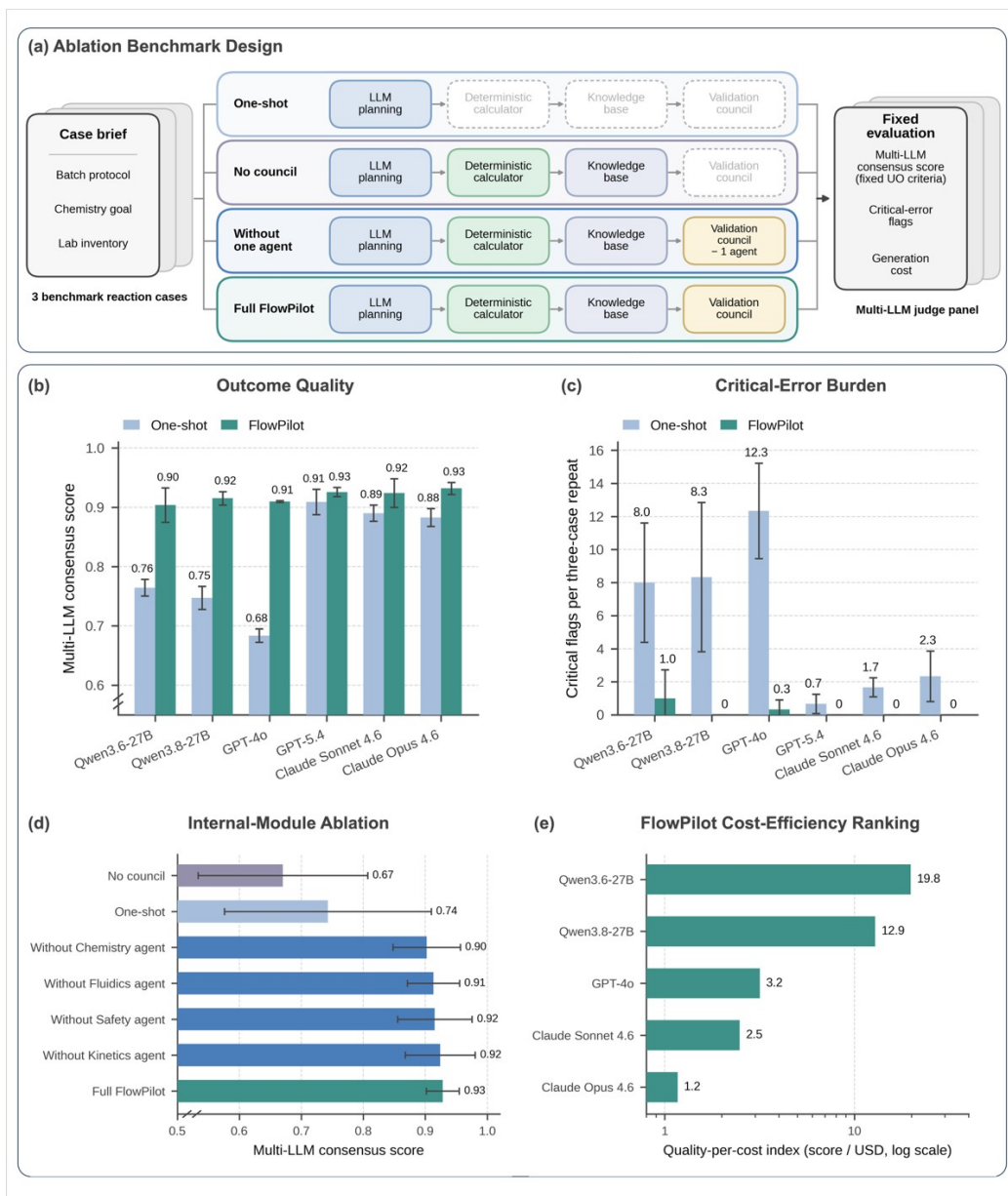


Figure 4. Ablation benchmarking and cost efficiency. (a) Architecture comparison under shared inputs and fixed evaluation. (b) Benchmark scores and (c) raw judge critical flags: mean and sample SD across three repeats, each containing three chemistry cases. The six-model display includes the archived GPT-5.4 generator comparison; this additional model is not included in the five-model aggregates in **Tables S11–S12**. (d) Selected Qwen3.8-27B ablations: mean and between-case SD with one generation per case-condition (all 15 conditions: **Table S10**). (e) Mean FlowPilot score divided by mean generation cost under the recorded price schedule, excluding judging and deployment. Critical flags are judge findings, not independently confirmed physical errors. ESI Section 3 provides evaluation details.

Across the five-model cohort in **Table S11**, full-pipeline mean scores ranged from 0.904 to 0.932, compared with 0.684 to 0.890 for matched one-shot generation (**Figure 4b**). The mean gain was largest for GPT-4o (+0.226), followed by Qwen3.8-27B (+0.168) and Qwen3.6-27B (+0.139); the Claude gains

were smaller (+0.034 and +0.049). The judges recorded 98 critical flags across the 45 one-shot outcomes and 4 across the 45 FlowPilot outcomes; 18 and 43 outcomes, respectively, had no critical flag (**Figure 4c**; **Table S12**). The score and flag summaries therefore point in the same direction within this cohort: improved completeness and fewer flagged inconsistencies. They do not establish universal model superiority or measured reaction success. **Figure 4d** provides a complementary internal comparison, with council removal causing the largest score decrease among the configurations shown.

The aggregate comparison is complemented by chemistry-specific scores (**Figure S7**), criterion-level score differences (**Figure S8**), and the location of critical flags within individual runs (**Figure S9**). **Figure S10** reports tokens, estimated generation costs, and observed runtime. The higher Qwen score-per-generation-cost values in **Figure 4e** therefore reflect the recorded token-price schedule, not a measurement of total deployment cost. **Tables S13-S14** define numerical closure and equipment checks; **Tables S15-S16** specify feedback and replay records. **Figure S11** and **Table S17** distinguish user-composed input from system-assembled prompts and returned artifacts for one matched run. This supports the practical input-handling argument without implying that FlowPilot uses fewer total model tokens. The legacy heuristic QA values in that worked example are not the fixed-criteria scores used in **Figure 4**.

The earlier council model-matrix and candidate-budget experiment is reported separately in **Figure S12** and **Tables S18-S21**. In the five-repeat comparison, the normalized radar-area score changed from 0.31 +/- 0.03 before council review to 0.48 +/- 0.23 afterward. This engineering-profile measure is not the multi-LLM benchmark score in **Figure 4**. The ESI defines Need revision, New versions, Combined pool, and Disqualified, and reports runtime, token use, geometry, and radar normalization. **Figure S13** and **Table S22** distinguish the final structured FlowPilot topology from its contemporaneous generated narrative; the complete one-shot response is retained alongside this worked example.

The ESI also shows how the chemist supplies and inspects the information behind a design. **Figures S14-S15** and **Table S23** document protocol entry and fixed-ID follow-up questions. **Figure S16** and **Table S24** show inventory import, normalization, and equipment warnings; **Figure S17** and **Table S25** show stage-resolved topology and stream parameters from a reopened run. Concrete failures are retained in **Table S26**, and **Figure S18** and **Table S27** expose actual council disagreement and selection records. **Figure S19** and **Table S28** place these views in the complete GUI and saved-run context. These operational records connect the computational benchmark to an inspectable workflow. The laboratory cases below examine how selected proposals translate into connected reactions and measured product yields.

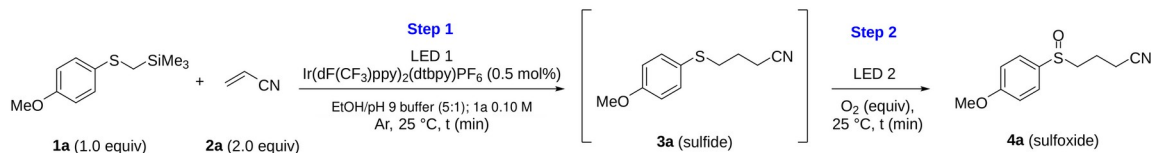
Case study 1: photoredox Giese addition followed by aerobic oxidation

The first connected-process case couples photoredox Giese addition with oxidation of the resulting sulfide to a sulfoxide (**Figure 5**). Photoinduced single-electron transfer generates an alpha-thiomethyl radical from an alpha-silyl sulfide; addition to acrylonitrile constructs the C–C bond, and subsequent oxygen-dependent photochemistry changes the sulfur oxidation state.⁵⁷ This sequence requires the oxygen-free first stage and oxidizing second stage to remain distinct within a connected process. The supplied design input specifies 4 h under argon followed by 6 h in air. These input durations are retained separately from the published batch comparison reproduced by KHU, which lists 180 and 360 min. Neither is treated as a kinetic law or converted using a universal intensification factor.

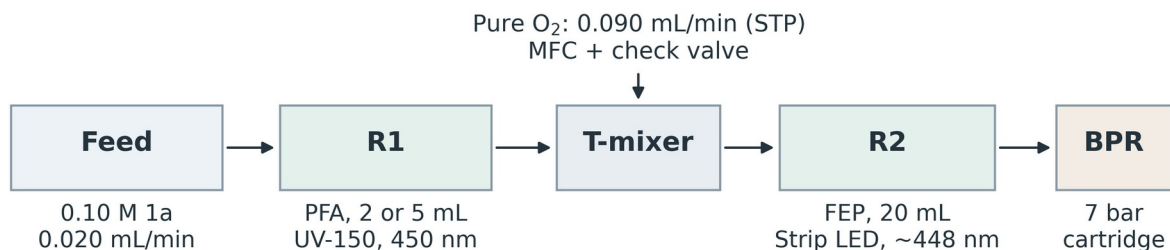
The revised proposal introduces pure oxygen only between the stages and binds each coil to a compatible irradiation module. KHU implemented a 2 or 5 mL PFA coil in the Vapourtec UV-150 module followed by a separately illuminated 20 mL FEP coil, both at 25 ± 1 °C. A 2.0 mL sample loop introduced a solution containing substrate 1a (0.100 M), acrylonitrile (0.200 M), and iridium photocatalyst (0.500 mM) into a carrier stream at 0.020 mL min⁻¹. Stage 1 therefore has a nominal liquid residence time of 100 or 250 min. The oxygen-controller setting was 0.090 mL min⁻¹. On the proposal's explicit STP basis (273.15 K, 1 atm), this corresponds to 0.00402 mmol min⁻¹ of oxygen and 2.01 equivalents relative to the nominal substrate molar flow. The laboratory MFC reference conditions require confirmation before this conversion is taken as a calibrated experimental value. The Stage 2 index, $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$, is 181.82 min. It combines reference-state gas volume with liquid flow and is not the residence time of either phase at operating pressure.

An earlier high-gas-flow trial showed upstream gas entry, prompting the move from air-based delivery to pure oxygen and a lower total gas flow. This change reduces the carrier-gas burden at a specified oxygen molar feed; it does not by itself establish hydraulic protection. The proposal includes a gas-line check valve, and the KHU photograph identifies a check valve in the assembled apparatus. Exact placement and startup-pressure behavior remain to be documented. KHU used a 7 bar cartridge BPR but recorded system pressures of 5.1–5.6 bar; the hardware rating and measured pressure are reported separately. The 2 mL first-stage configuration, grouped as Sets 1/3 in the supplied record, afforded sulfoxide 4a in 97% NMR yield and 94% isolated yield. Increasing the first-stage volume to 5 mL (Set 2) gave 98% NMR yield. Without replicate statistics, the one-percentage-point difference does not establish a benefit from the longer first stage. **Figure S20** and **Tables S29–S31** retain the design inputs and proposed settings; ESI Section 12, **Figures S23–S25** and **Tables S37–S39** document irradiation, implementation, and characterization. These measurements demonstrate productive telescoping, not a measured residence-time distribution or an independently optimized operating point.

a Connected chemical sequence



b FlowPilot proposal: connected reactor train



Ar-prepared feed; O₂ enters only after R1. Both stages: 25 °C.

c Reported operation and nominal stage times

Response set(s)	R1 / R2 (mL)	R1 time (min)	R2 index* (min)	Recorded pressure (bar)
1 and 3	2 / 20	100	181.8	5.1–5.4
2	5 / 20	250	181.8	5.2–5.6

d Measured sulfoxide 4a yields

Response set(s)	NMR yield (%)	Isolated yield (%)
1 and 3	97	94
2	98	Not reported

*R2 index = 20/(0.020 + 0.090), not operating-pressure residence time.

Shared-set entries are reported as supplied; experimental repeat counts were not provided.

Figure 5. Giese addition/oxidation: proposed configuration and measured outcomes. (a) Chemical sequence. (b) Simplified FlowPilot proposal (full topologies: **Figure S20**). (c) Implemented volumes and recorded pressures. R1 time is V/Q_{liquid} ; R2 uses the nominal inlet/STP index $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$, not operating-pressure residence time. MFC reference conditions require confirmation. The 7 bar cartridge rating is distinct from recorded pressure. (d) NMR and isolated yields supplied by KHU. Sets 1/3 share one reported entry, not documented independent repeats. Actual sample-loop operation, photographs, procedures, and analytical data are in ESI Section 12.

Case study 2: telescoped DPDTC-mediated amidation

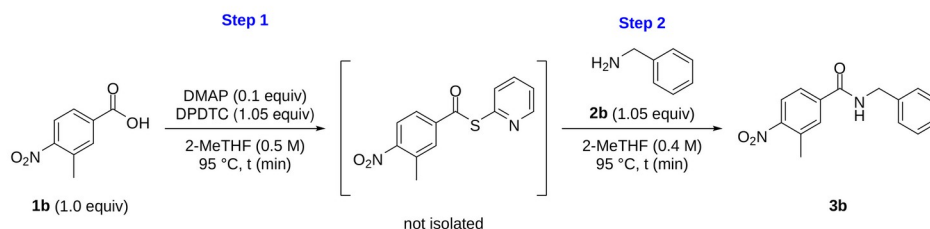
The second case is a thermal two-stage amidation of 3-methyl-4-nitrobenzoic acid with benzylamine (**Figure 6**). DPDTC and DMAP first generate a 2-pyridyl thioester; downstream aminolysis forms the amide C(O)–N bond while displacing the sulfur-containing leaving group.⁵⁸ The input specifies 30 min at 95 °C for each batch stage, separated by cooling and amine addition. In flow, the thioester is neither isolated nor assumed to form quantitatively. Stage 1 conversion limits the intermediate entering Stage 2, making final amide yield a response of the complete sequence rather than of either reactor alone.

KHU implemented the proposed separate-feed arrangement using two channels of the Vapourtec E-series system. Feed A contained acid (0.500 M), DPDTC (0.525 M), and DMAP (0.0500 M) in 2-MeTHF; Feed B contained benzylamine (2.10 M). In Set 1, $Q_A = 0.160$ and $Q_B = 0.0400$ mL min⁻¹ correspond to nominal acid and amine molar flows of 0.0800 and 0.0840 mmol min⁻¹, preserving 1.05 equivalents of amine. Two 5 mL ETFE coils give nominal stage times of 31.25 and 25.00 min because the second reactor receives both feeds. The reported implementation used a 2.0 mL sample loop for Feed A, direct interstage addition, and a 95 °C water bath for both coils. A 7 bar cartridge BPR was fitted; the recorded pressure for Set 1 was 5.2–5.3 bar. Thus hardware selection, nominal stage times, and measured pressure remain distinct parts of the process record.

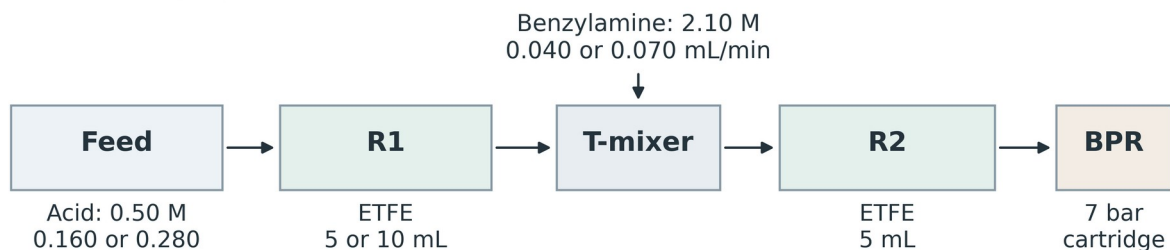
Set 1 afforded amide 3b in 86% NMR yield and 84% isolated yield. Sets 2/3 share a second reported operating point: a 10 mL first reactor and 5 mL second reactor with feeds of 0.280 and 0.0700 mL min⁻¹, giving nominal stage times of 35.71 and 14.29 min and 68% NMR yield. The nominal combined stage time decreases from 56.25 to 50.00 min while the substrate molar feed rises from 0.0800 to 0.140 mmol min⁻¹. The lower final yield despite a longer first stage is consistent with a coupled yield–throughput trade-off, but cannot identify the limiting step because both flow and reactor configuration changed and intermediate conversion was not measured. **Figure S21** and **Tables S32–S34** retain the response sets and design calculations. **Figures S26–S27** provide the supplied experimental scheme and apparatus photograph; **Tables S38–S39** reconcile implementation and molar feeds. The measurements support intermediate transfer without isolation while showing why numerical closure alone does not determine the best chemical operating point.

The alpha-bromination example from the original preprint⁵⁹ is retained unchanged as **Figure S22** in ESI Section 9 for historical comparison. It is distinct from the current amidation case and is not presented as a new laboratory validation.

a Connected chemical sequence



b FlowPilot proposal: connected reactor train



Flows: mL/min. Both stages: 95 °C; no intermediate isolation.

c Reported operation and nominal stage times

Response set(s)	Q_A / Q_B (mL/min)	R1 / R2 (mL)	R1 / R2 time (min)	Recorded pressure (bar)
1	0.160 / 0.040	5 / 5	31.25 / 25.00	5.2-5.3
2 and 3	0.280 / 0.070	10 / 5	35.71 / 14.29	5.5-5.8

d Measured amide 3b yields

Response set(s)	NMR yield (%)	Isolated yield (%)
1	86	84
2 and 3	68	Not reported

$R1 \text{ time} = V_1/Q_A$; $R2 \text{ time} = V_2/(Q_A + Q_B)$.

Shared-set entries are reported as supplied; experimental repeat counts were not provided.

Figure 6. DPDTc-mediated telescoped amidation: proposal and laboratory implementation. (a) Acid activation to a 2-pyridyl thioester followed by benzylamine addition and amide formation. (b) Simplified proposed connected train; complete archived topologies are in **Figure S21**. (c) Implemented flow rates, reactor volumes, nominal liquid residence times, and KHU-reported system pressures. R1 uses V_1/Q_A , whereas R2 uses $V_2/(Q_A + Q_B)$. Both stages were held at 95 °C; the 7 bar cartridge rating is distinct from recorded pressure. (d) Reported NMR and isolated yields. Sets 2/3 share one experimental entry; independent replicate statistics were not supplied. Sample-loop operation, detailed preparation, photographs, and analytical records are in ESI Section 12.

DISCUSSION

FlowPilot addresses batch-to-flow translation as a connected-process design task. Chemistry interpretation, literature retrieval and specialist review determine what should be proposed, while deterministic calculations reconcile dependent quantities and equipment assignments. Their contributions are evaluated at different levels: **Figure 3** measures retrieval alignment, **Figure 4** compares delivered designs, and the component ablations examine the role of review. A shared process record connects these steps to the stream table and topology, making the relationship between chemical assumptions and engineering conditions explicit.

The matched architecture comparison shows higher mean scores with FlowPilot for each of the five retained generators, with the improvement depending on the model (**Figure 4b**). In the internal Qwen3.8-27B screen, removing the council reduced the mean score from 0.928 to 0.670 across the three cases (**Figure 4d**; **Table S10**). Individual specialist removals had smaller effects. This single-generation-per-cell screen is a descriptive component comparison, not a statistical demonstration that every specialist is necessary. The historical model-matrix and budget experiments further show that broader candidate exploration does not guarantee better outcomes (**Figure S12**; **Tables S18-S21**). Together, these results support coordinated review while showing why configuration matters.

The connection between chemical steps, quantitative feeds, and available equipment is central to this approach. It complements microreactor-design assistants, experimentally integrated synthesis platforms, and simulator-based process agents by keeping the proposed laboratory configuration and its dependent calculations in one record.⁴⁶⁻⁵¹ The same emphasis on explicit process state can support subsequent optimization and materials-development workflows.^{60,61}

The archived examples show why record-level inspection remains important alongside aggregate scores. **Table S26** includes conflicting one-shot residence-time bookkeeping and a FlowPilot catalyst-preparation inconsistency, while **Table S22** documents disagreement between structured conditions and an accompanying narrative. These examples motivate a common final record and traceable revision decisions. Retaining the inputs, calculations, equipment selections and council exchanges makes both design conflicts and evaluator errors available for examination rather than hiding them behind a single score.

The laboratory cases connect this computational assessment to chemical outcomes. Giese addition/oxidation demonstrates a productive sequence with oxygen introduced downstream, while the closely similar yields at two first-stage volumes do not establish an optimum. DPDTC amidation

demonstrates intermediate transfer without isolation, but its lower yield at higher feed rate shows that a numerically closed proposal need not be the chemically preferred operating point. Both were implemented using sample-loop injections; extended steady-state operation and repeatability are separate questions. The feedback interface can incorporate measured yields together with the actual settings and deviations, rather than treating a proposed design as experimental evidence. Integration with Bayesian optimization³⁰⁻³⁴ and language-guided priors⁶² offers a route to test chemist hypotheses against such accumulated records.

CONCLUSION

FlowPilot connects a batch protocol and chemist-supplied constraints to a traceable first flow-process proposal. Standardized intake establishes the design context, chemistry-aware retrieval supplies precedents, deterministic calculation links dependent parameters, and specialist review assesses candidate operating choices. Equipment reconciliation and a common final record connect the selected conditions to the process diagram and supporting report.

For the five-model, three-chemistry cohort, the mean fixed-criteria score increased from 0.794 for one-shot generation to 0.917 for FlowPilot, with fewer raw judge critical flags. Module ablations support a contribution from council review, while historical model-matrix and budget studies show configuration sensitivity. Laboratory implementation produced a sulfoxide in 94% isolated yield and an amide in 84% isolated yield under selected connected-stage conditions. The different amidation outcomes at two operating points reinforce the need to evaluate chemical performance separately from computational consistency. Together, these findings support a traceable workflow from reaction-level input to an experimentally reviewable flow configuration within the scope of the cases examined.

For the chemist, the intended benefit is a shorter path from an initial procedure to an inspectable proposal: enter the protocol, resolve the standardized questions, select an inventory profile, and review the linked design records. Measurements can then be returned with the actual conditions to inform a subsequent design cycle. This connects proposal generation with experimental development while preserving the distinction between what was assumed, calculated, proposed, and observed.

More broadly, the hybrid pattern demonstrated here, LLM-driven interpretation coupled with deterministic computation and multi-agent cross-domain audit, is not specific to flow chemistry. Any engineering design problem that is currently solved by a small team of complementary specialists iterating across coupled domains is a candidate for the same architectural approach. The modular design of FlowPilot is intended to make such extensions practical.

METHODS

Standardized inputs and study organization

Each design began with a batch protocol and a structured intake package recording the objective, chemistry identity, experimental history, laboratory inventory, operating limits, and chemist hypotheses. Follow-up questions used stable identifiers and predefined readiness rules; optional evidence could be explicitly marked unavailable. The confirmed package was frozen for the run, retaining the answers and their provenance. Measured observations, protocol facts, and hypotheses were stored separately, while declared equipment and operating limits constrained design feasibility. Computational comparisons used archived case packages rather than newly generated conversations, keeping the supplied chemistry and constraints constant between architectures. The intake contract and question rules are given in ESI Section 1 and **Tables S1–S3**. Benchmark campaigns and subsequent laboratory-design runs were treated as distinct software/configuration records; the dated development history is summarized in **Table S35**.

Literature corpus and plan-aware retrieval

The literature resource comprised 464 classified flow-chemistry records annotated with available reaction, reactor, material, operating-condition, yield, and inlet-stream information. Missing metadata were retained as missing rather than filled from model inference. A complementary handbook-derived store contained 2,537 engineering rules with topic, severity, source, and applicability information.^{3,4,21} Relevant rules supplied context to chemistry interpretation; only checks implemented in the calculation and validation routines acted as deterministic constraints. Field completeness and rule coverage are reported in **Tables S4–S5**.

A chemistry plan was extracted from the intake package and used to construct a retrieval query from reaction class, mechanism, catalyst, solvent, phase regime, and operating conditions. The search began with paired batch-flow records under mechanism and phase filters. Filters were relaxed when fewer than three candidates remained, and the search expanded to all records if no paired match remained. Reranking combined 0.60 semantic similarity with 0.40 field similarity; the latter compared photocatalyst, solvent, wavelength, temperature, and concentration using the weights in **Table S6**. The designated source record for each benchmark was excluded before final selection. This controlled direct retrieval of the reference, not possible exposure during model pretraining. Retrieval was evaluated by rank changes and top-five photocatalyst-family matching against semantic search over the frozen 1,600 query-result pairs (**Table S7**; ESI Section 2).

Engineering calculations and stage accounting

The engineering layer recalculated quantities dependent on proposed operating choices. Component molar delivery was obtained from concentration and liquid volumetric flow, and each stage received the cumulative flow of the streams introduced at or upstream of that stage. For a nominal liquid-filled reactor, the stage time was calculated as reactor volume divided by local liquid flow. An interstage feed therefore changed the downstream material balance and time calculation without changing the upstream flow. Packed-bed empty volume, estimated liquid holdup, and catalyst-contact time were distinguished where applicable. Stage totals were formed only from times reported on compatible bases.

Gas equivalents were calculated from the reactive-gas molar flow divided by the limiting-reactant molar flow, including gas composition and the stated reference temperature and pressure. The proposal STP convention was 273.15 K and 1 atm. Where required by a benchmark contract, in-channel gas volume was derived by ideal-gas scaling using absolute operating pressure and temperature. The inlet-reference index $V/(Q_{\text{liquid}} + Q_{\text{gas,STP}})$ was kept separate from pressure-corrected apparent time and from an experimentally measured residence time. Pressure-loss and thermal screening used the selected geometry, phase assumptions, and available material properties; estimated heat-transfer or reaction-enthalpy inputs were identified as assumptions. The implemented quantities, multiphase conventions, and heat-transfer screening are detailed in ESI Section 4 and **Table S13**.

Candidate generation and council review

The downstream Designer generated alternative geometries and operating choices from the chemistry plan, retrieved analogies, engineering calculations, and inventory constraints. The Chemistry, Kinetics, Fluidics, and Safety specialists reviewed chemical compatibility, reaction-time plausibility, transport and hardware feasibility, and hazards, respectively. The Skeptic examined assumptions and contradictions, and the Orchestrator selected among the reviewed candidates. Agent verdicts and proposed revisions were stored with candidate identifiers. Changes to independent variables triggered deterministic recalculation before a candidate could be compared or published. Candidate and revision budgets followed the configuration recorded for each campaign rather than being inferred from the final output. Rejected alternatives and revision history were retained alongside the selected candidate. Council terminology, configuration studies, and examples of actual deliberation are documented in ESI Sections 6.1 and 7.5.

Inventory realization and result publication

Inventory profiles were imported as structured data or normalized from laboratory documents into typed equipment records with stable identifiers. Available quantities, system membership, material, reactor

volume and diameter, pump range, pressure and temperature limits, and compatible light modules were recorded where supplied. Realization assigned required operations to available items and checked the declared capability and connectivity constraints. Calculations were repeated when equipment selection changed a dependent quantity. An unresolved essential requirement was retained as a diagnostic requirement, not silently replaced with invented hardware (**Table S14**).

The final design contract reconciled the selected proposal, stage table, stream table, equipment allocation, and topology. GUI numerical views and the executable process diagram were derived from that record; earlier calculations and council exchanges remained separately available for inspection. Timestamped run folders stored the input package, result, final contract, topology, equipment allocation, and available diagram artifacts. Experimental feedback was linked to the conditions actually tested before being supplied to a later design cycle. ESI Sections 1.4–1.5 and 7 document the GUI and saved artifacts, and **Tables S15–S16** specify the feedback and provenance records. Exact computational replay additionally requires the model, decoding settings, dependencies, and software revision recorded for the campaign.

Matched architecture comparison and independent judging

The primary comparison used CuAAC, photochemical oxidation, and hydrogenolysis case packages with the same protocol, objective, inventory, and final-output contract for both architectures. One-shot generation received this information in a single call; FlowPilot distributed it across its interpretation, calculation, review, and realization stages. Qwen3.6-27B, Qwen3.8-27B, GPT-4o, Claude Sonnet 4.6, and Claude Opus 4.6 were each used with both architectures for three generation repeats per case, giving 90 outcomes at the recorded generation temperature of 0.2. The archived GPT-5.4 generator results additionally displayed in **Figure 4b–c** were not included in these five-model aggregates or **Tables S11–S12**. Exact case prompts, inventories, and model records are retained in ESI Section 3. The comparison concerns delivered design quality under matched task information, not equal token or computation budgets.

For judging, final designs were normalized into candidate packets with generator and architecture labels removed. Separate Qwen, OpenAI, and Claude judge calls received the candidate, common input context, reference facts, deterministic verification evidence, and the fixed 14-criterion rubric. The archived judge configuration used Qwen3.6-27B, GPT-5.4, and Claude Sonnet 4.6 at temperature 0.0; these evaluator calls were separate from candidate generation. Each applicable criterion received an integer rating from 0 to 4. Ratings were averaged across judges for each criterion, then across applicable criteria, and divided by four to give a score from 0 to 1. All judges and applicable criteria had equal weight. Gas bookkeeping and multistage closure could be excluded only when their applicability conditions were false; missing

required information was scored rather than treated as inapplicable. Critical flags were counted separately from the numerical score. Deterministic evidence did not confer an architecture-specific bonus, and identity masking did not guarantee that output style concealed the architecture. Scoring anchors, criterion definitions, and applicability rules are specified in **Tables S8–S9**.

Ablation experiments and statistical reporting

For each generator and architecture, the three case scores were averaged within each repeat. Reported architecture error bars are the sample standard deviation of these three repeat-level means, not the dispersion of all nine individual outcomes. Critical-flag counts were summed across the three cases within a repeat before calculating their mean and sample standard deviation. The aggregate values in **Tables S11–S12** use only the five-model cohort. Generation cost was estimated from recorded token usage and the archived model-specific price schedule, excluding evaluator calls and deployment costs. **Figure 4e** uses mean score divided by mean generation cost; the case-specific ESI plots instead summarize per-outcome score-to-cost ratios (**Figure S10**).

The internal ablation screen fixed the generator to Qwen3.8-27B and tested 15 conditions over the same three cases, producing 45 outcomes with one generation per condition-case pair. The conditions removed or modified council components, revision stages, selection, or candidate budgets. Disabled specialists received no substitute score, and active council weights were renormalized. The resulting standard deviations describe between-case variation, not repeatability; all conditions are reported in **Table S10**. A separate historical study crossed four upstream and four council endpoints and tested candidate budgets of 1, 6, 12, and 24 with five repeats. Its engineering-profile metrics were normalized independently of the LLM-judge rubric (ESI Section 6.1; **Tables S18–S21**). No experimental yield uncertainty was inferred from these computational repeats.

Laboratory implementation and chemical analysis

KHU implemented the two connected-stage cases using a Vapourtec E-series system with sample-loop injection. For the Giese addition/oxidation sequence, a PFA coil in a 450 nm UV-150 module was followed by a separately illuminated FEP coil. The substrate/acrylonitrile/iridium-catalyst feed was prepared for the oxygen-free first stage; pure oxygen was introduced through an MFC at the interstage mixer, and both reactors were maintained at 25 ± 1 °C. For DPDTC-mediated amidation, the acid/DPDTC/DMAP solution and benzylamine solution were delivered by separate channels, with amine addition between two ETFE coils maintained at 95 °C. Both setups used a 7 bar cartridge BPR; recorded system pressure was reported separately from the cartridge rating. Actual concentrations, molar feeds,

operating points, and proposal-to-implementation differences are tabulated in ESI Section 12 and **Tables S31, S34, and S38–S39**.

NMR yields were reported using 1,3-benzodioxole as the internal standard. Isolated products were characterized by ^1H and ^{13}C NMR spectroscopy and low-resolution mass spectrometry. Shared response-set entries in the supplied laboratory record were treated as grouped operating points, not independent replicates, and no experimental standard deviation was assigned without individual measurements. ESI Section 12 contains the feed preparation, work-up, irradiation specifications, apparatus photographs, and characterization records; **Figures S28–S31** reproduce the product spectra. MFC reference-state confirmation, collection windows, and quantitative-NMR calculation records are identified there as outstanding metadata rather than filled by inference.

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Notes

Availability of data and code

Source code is publicly accessible at <https://github.com/armtfgh/flowpilot>. Benchmark scripts and the data-access index are available at https://github.com/armtfgh/flowpilot/tree/master/ablation_test and https://github.com/armtfgh/flowpilot/tree/master/ablation_results, respectively. The latter is currently an index, not the complete outcome archive. The ESI contains the reported numerical summaries, experimental procedures, and characterization spectra. As of 29 September 2026, the repository is public but has no tagged manuscript release; the complete benchmark archive, raw experimental analytical files, and a persistent data DOI have not yet been deposited. These release items remain to be completed before submission.

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ABBREVIATIONS

BPR, back-pressure regulator; DPDTC, dipyridyldithiocarbonate; DMAP, 4-(dimethylamino)pyridine; Da, Damköhler number; ETFE, ethylene tetrafluoroethylene; FEP, fluorinated ethylene propylene; IF, intensification factor; LLM, large language model; LRMS, low-resolution mass spectrometry; MFC, mass flow controller; 2-MeTHF, 2-methyltetrahydrofuran; NMR, nuclear magnetic resonance; Pe, Péclet number; PFA, perfluoroalkoxy; PFD, process flow diagram; PTFE, polytetrafluoroethylene; RAG, retrieval-augmented generation; Re, Reynolds number; STP, standard temperature and pressure (273.15 K and 1 atm for the proposal calculations); STY, space-time yield.

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